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RESEARCH ARTICLE

No intrinsic number bias: Evaluating the role of perceptual discriminability in magnitude categorization

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Abstract

Accumulating evidence suggests that there is a spontaneous preference for numerical, compared to non-numerical (e.g., cumulative surface area), information. However, given a paucity of research on the perception of non-numerical magnitudes, it is unclear whether this preference reflects a specific bias towards number, or a general bias towards the more perceptually discriminable dimension (i.e., number). Here, we found that when the number and area of visual dot displays were matched in mathematical ratio, number was more perceptually discriminable than area in both adults and children. Moreover, both adults and children preferentially categorized these ratio-matched stimuli based on number, consistent with previous work. However, when number and area were matched in perceptual discriminability, a different pattern of results emerged. In particular, children preferentially categorized stimuli based on area, suggesting that children's previously observed number bias may be due to a mismatch in the perceptual discriminability of number and area, not an intrinsic salience of number. Interestingly, adults continued to categorize the displays on the basis of number. Altogether, these findings suggest a dominant role for area during childhood, refuting the claim that number is inherently and uniquely salient. Yet they also reveal an increased salience of number that emerges over development. Potential explanations for this developmental shift are discussed.

KEYWORDS
area perception, categorization, non-symbolic number, numerical cognition, vision

Research Highlights

- Previous work found that children and adults spontaneously categorized dot array stimuli by number, over other magnitudes (e.g., area), suggesting number is uniquely salient.
- However, here we found that when number and area were matched by ratio, as in prior work, number was significantly more perceptually discriminable than area.

- When number and area were made equally discriminable ('perceptually-matched'), children, contra adults, spontaneously categorized stimuli by area over number (and other non-numerical magnitudes).
- These findings suggest that area may be uniquely salient early in childhood, with the previously-observed number bias not emerging until later in development.

1 | INTRODUCTION

Many theories have been put forth to explain how humans represent numerical quantity, with the dominant view arguing that even non-symbolic number representations are abstract. Consistent with this possibility is research demonstrating that pre-verbal humans and nonhuman animals represent number from different modalities in a common format (Anobile et al., 2016; Izard et al., 2009; Jordan et al., 2008). Some researchers who ascribe to this view argue further that number is perceived directly and independently of other magnitudes. That is, number representations are not derived from non-numerical magnitudes, such as cumulative surface area or density (Anobile et al., 2019; Clarke & Beck, 2021; Park et al., 2016). Most recently, some have also argued that number is an intrinsically and uniquely salient quantitative dimension, such that it is favored over other magnitudes in behavioral judgments even when more general factors such as stimulus variability are accounted for (Cicchini et al., 2016; Tomlinson et al., 2020). Although the existence of abstract representations of non-symbolic number (i.e., 'number sense') does not necessitate these latter claims (i.e., number as perceived directly and uniquely salient), these claims nonetheless are commonly leveraged as support for the existence of abstract number representations (Anobile et al., 2016; Dehaene, 2011; Odic & Starr, 2018). Thus, in the remainder of this manuscript, we refer to this suite of claims as the 'Number Sense' account.

An alternative perspective, referred to as the 'Magnitude Sense' account, argues that number is not uniquely salient (Henik et al., 2012, 2017; Leibovich & Henik, 2013; Leibovich et al., 2017). From this perspective, the claim is that numerical abilities develop from a more fundamental ability to perceive and discriminate 'continuous magnitude', such as cumulative surface area (i.e., the total area of a set of objects; referred to from here on as 'area'). Though many researchers agree that numerical abilities may have evolved from non-numerical magnitude representations (Agrillo & Bisazza, 2018; Amalric & Dehaene, 2016; Cantlon et al., 2009), Henik et al. (2012) further posited that human development recapitulates this evolutionary process. Specifically, Leibovich, Henik and colleagues have suggested that numerical abilities are not present at birth. Instead, numerical abilities emerge as a consequence of children's capacity for statistical learning, which allows for recognition of the correlation between numerical and non-numerical magnitudes in the physical environment (Gebuis et al., 2016). By late childhood, the acquisition of symbolic/linguistic representations of number (e.g., number words) further orients attention towards numerical information (Leibovich et al., 2017; Newcombe et al., 2015). Accordingly, this perspective argues that prior to exposure to symbolic number and formal mathematics, non-numerical magnitudes, such as area, element size, and contour length, are more salient than number (Clearfield & Mix, 1999; Newcombe, 2014; Viarouge et al., 2018).

To assess whether number is uniquely (and universally) salient, Ferrigno et al. (2017) examined behavior from American adults and children, Tsimané adults (who lack experience with symbolic number), and rhesus macaques on a magnitude categorization task (see also, Cantlon & Brannon, 2007). Participants were instructed to categorize dot arrays, which varied parametrically in number and area into two categories. During the training phase, participants were given feedback (correct or incorrect) in order to learn that one category corresponded to 'small' and the other corresponded to 'large' (though participants were not given any category labels). Number and area were (purposefully) correlated in the training stimuli, such that participants could correctly categorize stimuli based on number, area, or both. During the test phase, number and area were uncorrelated, thus revealing what 'rule' participants encoded during training (i.e., number, area, or both). If number is uniquely salient, as the authors predicted, then all groups should exhibit a bias toward categorizing stimuli based on numerical value (i.e., one category for numerically small stimuli and one category for numerically large stimuli), despite differences in area. Indeed, the authors found that all groups preferentially categorized the stimuli by number, with American adults exhibiting the strongest preference for number of all four groups. Consistent with the Number Sense account, the findings of Ferrigno et al. (2017) suggested that despite the presence of other visual magnitude dimensions in the physical environment, number is spontaneously perceived and uniquely salient (see also, Cordes & Brannon, 2009; Tomlinson et al., 2020). The ubiquity of this number bias is especially striking, as it suggests that exposure to human culture, language, and symbolic number systems may not be necessary for the development of such a bias (but see, Agrillo et al., 2016).

A key strength of the Ferrigno et al. (2017) study was that they examined magnitude salience with a spontaneous categorization task (see also, Viarouge et al., 2018, 2019). Not only did this task not require any explicit judgment of the number or area of the dot arrays, but also, task instructions did not reference the dimensions of interest (i.e., number and/or area). Despite these strengths, it is unclear what role potential differences in perceptual discriminability may have played in these, and others', findings (Cicchini et al., 2016; Hurewitz et al., 2006).

Specifically, it is unclear whether the putative number bias reflects a unique and primary role of numerical information in categorization or, instead, a difference in discriminability, such that number was more discriminable in all participant groups and, consequently, used for categorizing stimuli.

Indeed, decades of work on perceptual learning, categorization, and psychophysics have shown that category judgments tend to align with the most perceptually variable dimension (Goldstone, 1994; Jones & Goldstone, 2013). Critically, the perceptual variability of a particular stimulus dimension depends both on the range of stimulus values presented, as well as the discriminability of that dimension (Melara & Mounts, 1994). Thus, in order to determine whether one dimension is more *intrinsically* salient than another, it is critical that dimensions be matched on perceptual discriminability. Perceptual discriminability is considered matched when values in one dimension are as psychologically (i.e., perceptually) different as values in the other dimension (Melara & Mounts, 1993).

When comparing the perception of numerical and non-numerical magnitudes, researchers have typically equated stimulus values with respect to mathematical ratio (de Hevia et al., 2012; Ferrigno et al., 2017; Starr & Brannon, 2015). This norm is not unfounded; extensive work in numerical cognition has shown perceptual discriminability for number is better explained by the ratio between stimuli than by the absolute difference, as predicted by Weber's law (Dehaene, 2003). For example, it is equally difficult to discriminate 8 versus 10 dots as it is to discriminate 80 versus 100 dots because they reflect the same ratio. However, and critically, the validity of this ratio-matching strategy for stimulus control relies on an unfounded assumption: that matching dimensions in their mathematical variability necessarily matches dimensions in their perceptual variability (Barth, 2008). Indeed, the robust finding that ratio determines discriminability *within* a dimension does not imply that ratio determines perceptual discriminability *across* dimensions. For example, it is unclear whether the difference between 8 and 10 dots (number) and 8 and 10 cm^2 (area), which is mathematically equivalent (8 vs. 10i n both cases), is perceptually-matched. Missing from this literature is a key insight from psychophysics (Goldstone, 1998; Shepard, 1981)—namely, the relation between physical difference and psychological difference is not one-to-one.

Both children and adults tend to have greater acuity for numerical information compared to area (Cicchini et al., 2016; Cordes & Brannon, 2008; Tomlinson et al., 2020; cf. Odic et al., 2013). This differential acuity for number and area suggests that, even when matched for mathematical ratio, both children and adults find number more perceptually discriminable than area, and thus, more perceptually variable. In other words, all else being equal, stimuli that differ by a 2:1 ratio in number (e.g., 16 and 8 dots) will seem more different than stimuli that differ by a 2:1 ratio in area (e.g., 16and 8 cm^2). Accordingly, this difference in perceptual variability could account for a bias towards number, as opposed to area, without invoking any difference in the inherent salience of the two dimensions.

The conflation between acuity and salience also jeopardizes the interpretation of measures of magnitude interference, such as congruency effects (see Barth, 2008). For example, in work using magnitude comparison tasks, a magnitude dimension is considered uniquely salient if it interferes with judgments of other magnitudes more than other magnitudes interfere with judgments of it. Similarly, in work using looking-time paradigms with infants, a magnitude dimension is considered uniquely salient if infants preferentially attend to, or discriminate based on, changes in that magnitude even when it is 'pitted against' other magnitudes (Clearfield & Mix, 1999). For example, infants attended to a fourfold increase in number, even when those increases were accompanied by a simultaneous fourfold decrease in area (Cordes & Brannon, 2009).

However, it is possible that differential interference between magnitudes does not (exclusively) reflect unique properties of the magnitude dimensions, but rather, reflects differences in acuity for those dimensions. For example, Barth (2008) found that congruency effects, wherein area judgments were facilitated by congruent number information, resulted not from an interaction between magnitude dimensions, but from a discrepancy between veridical (i.e., mathematical) and perceived area. Drawing from previous work showing that adult participants commonly underestimate area (Chong & Treisman, 2003; Teghtsoonian, 1965), Barth found that when area values were calculated based on perceived (i.e., underestimated) area, as opposed to mathematical area, incongruent trials (i.e., trials where the stimulus that was larger in area was smaller in number) contained systematically smaller area differences than congruent trials. Moreover, and critically, differences in participants' performance between incongruent and congruent trials was better explained by the area ratios than by the degree of congruency between number and area. In other words, better performance on congruent, compared to incongruent, trials, was not due to facilitation from number but simply because congruent trials contained larger area ratios, and thus, were easier to discriminate. These findings suggest that, because mathematical area itself does not adequately describe participants' ability to perceive and discriminate area, it is inappropriate to measure perceptual interactions between magnitudes with respect to these mathematical area values.

Aulet and Lourenco (2021a)'s reanalysis of Tomlinson et al. (2020) likewise demonstrated the importance of considering differences between mathematical and perceived area. Tomlinson et al. found that for both adults and children (five-year-olds), number influenced participants' area judgments more than area influenced their number judgments. Based on this asymmetric interference, Tomlinson et al. argued that number is uniquely salient. By contrast, Aulet and Lourenco found that when differences in perceptual discriminability between number and area were accounted for, by recalculating area values for the stimuli based on an alternative model of area perception (i.e., 'additive area'; Yousif & Keil, 2019), area was more salient than number early in development. Specifically, during an explicit magnitude comparison task, children's judgments of number were influenced by area more than their judgments of area were influenced by number.

However, the findings of Barth (2008), Tomlinson et al. (2020), and the subsequent reanalysis by Aulet and Lourenco (2021a), were based on explicit magnitude comparison tasks, in which participants were instructed to judge which of two displays was larger in number or area. Accordingly, the greater salience of number, at least in adults, could be due to decisional, post-perceptual, effects, perhaps facilitated by experience with symbolic number and formal mathematical education (Hollingworth, 1910; Petzschner et al., 2015; Poulton, 1979). In other words, the salience of number may not reflect anything *perceptually* unique about number but, instead, may be due to a learned, non-perceptual bias towards number, specifically within the context of magnitude estimation.

1.1 | Present study

Here, we evaluated the influence of perceptual discriminability on the relative salience of number and area during spontaneous categorization. If number is intrinsically salient, then a preference to categorize stimuli by number, compared to area, should persist even when stimuli are perceptually-matched. In contrast, if the previously observed number bias results from number being more discriminable than area, then a preference to categorize stimuli by number compared to area should no longer be observed. That is, when number and area are perceptually-matched, participants should categorize by number and area, or area alone. Moreover, if a number bias develops as a result of experience with symbolic number and mathematics, then children may exhibit less of a number bias than adults.

To this end, in the present work, we first assess whether number and area are equally discriminable in ratio-matched stimuli in adults and children (Experiments 1A and 2A), using a magnitude comparison task (see also, Aulet & Lourenco, 2021b; Castaldi et al., 2018). Following results suggesting that the stimuli were not equally discriminable in number and area, we generated new stimuli ('ratio-adjusted' stimuli) to account for observed differences in discriminability. We then assessed whether number and area were in fact equally discriminable in the ratio-adjusted stimuli in adults and children (Experiments 1B and 2B). With the ratio-matched and ratio-adjusted (subsequently referred to as 'perceptually-matched) stimuli, we then evaluated the influence of number and area on adults' and children's spontaneous category judgments (Experiments 3and 4, respectively), and assessed whether this influence differed as a function of stimulus condition (ratio-matched or perceptually-matched).

2 | EXPERIMENT 1: MAGNITUDE COMPARISON (ADULTS)

Though the primary goal of this work was to evaluate the salience of number and area during categorization, we first assessed the baseline discriminability of number and area in adults (Experiment 1) and children (Experiment 2). Importantly, previous research that has suggested number is more salient than area did not ensure that number and area were matched in baseline discriminability (e.g., Ferrigno et al., 2017; Tomlinson et al., 2020), a critical precondition for evaluating dimensional salience in multidimensional stimuli (Melara & Mounts, 1993, 1994). To this end, we used a magnitude comparison task, in which participants had to judge which, of two dot arrays, was greater in number or area, to evaluate the baseline discriminability of number and area (see also, Aulet & Lourenco, 2021b; Castaldi et al., 2018).

2.1 | Ratio-matched stimuli

In Experiment 1A, we evaluated the discriminability of number and area in adults with ratio-matched stimuli. If number and area are not equally discriminable when matched by ratio, then this would suggest that the unique salience of number could be a byproduct of this difference in discriminability.

2.1.1 | Methods

Participants

Eighteen undergraduates (M_{age} = 19.69 years) participated in this experiment for course credit. All participants had normal or corrected-to-normal vision. Procedures were approved by the local Institutional Review Board (IRB). All tasks were presented on a desktop computer with a 19-inch screen (1280 × 1024 px) located in an experimental laboratory. Participants sat approximately 60 cm from the computer screen.

Stimuli

Stimuli were dot arrays comprised of gray dots on a black background (400 × 400 px), varying parametrically in number and area. As in Ferrigno et al. (2017), number and area values were 8, 10, 12, 14, 16, 18, 20, and 22 dots or cm^2, respectively (see Figure 1a). Five exemplars were generated for each array (random dot placement), resulting in 320 unique stimuli per set (64 number/area values × 5 exemplars). On each trial, the particular exemplar used was chosen at random with replacement.

Procedure

Participants completed two blocks of magnitude comparison trials. In one block, participants indicated which of two dot arrays was greater in number; in the other block, participants indicated which of two dot arrays was greater in area (counterbalanced order[1]). Each block consisted of 500 trials. Trials were randomly selected, without replacement, from all possible stimulus pairs in which the stimuli differed in both number and area ($n = 3136$).

Stimuli were presented on the center left and center right of the screen (side counterbalanced across trials), for 1500 ms. Participants had unlimited time to respond. Participants were instructed to press the "q" key if the left array was greater in number (or area) and the "p" key if the right array was greater in number (or area).

2.1.2 | Results

To determine whether number and area were equally discriminable, we evaluated participants' accuracy on trials in which adjacent

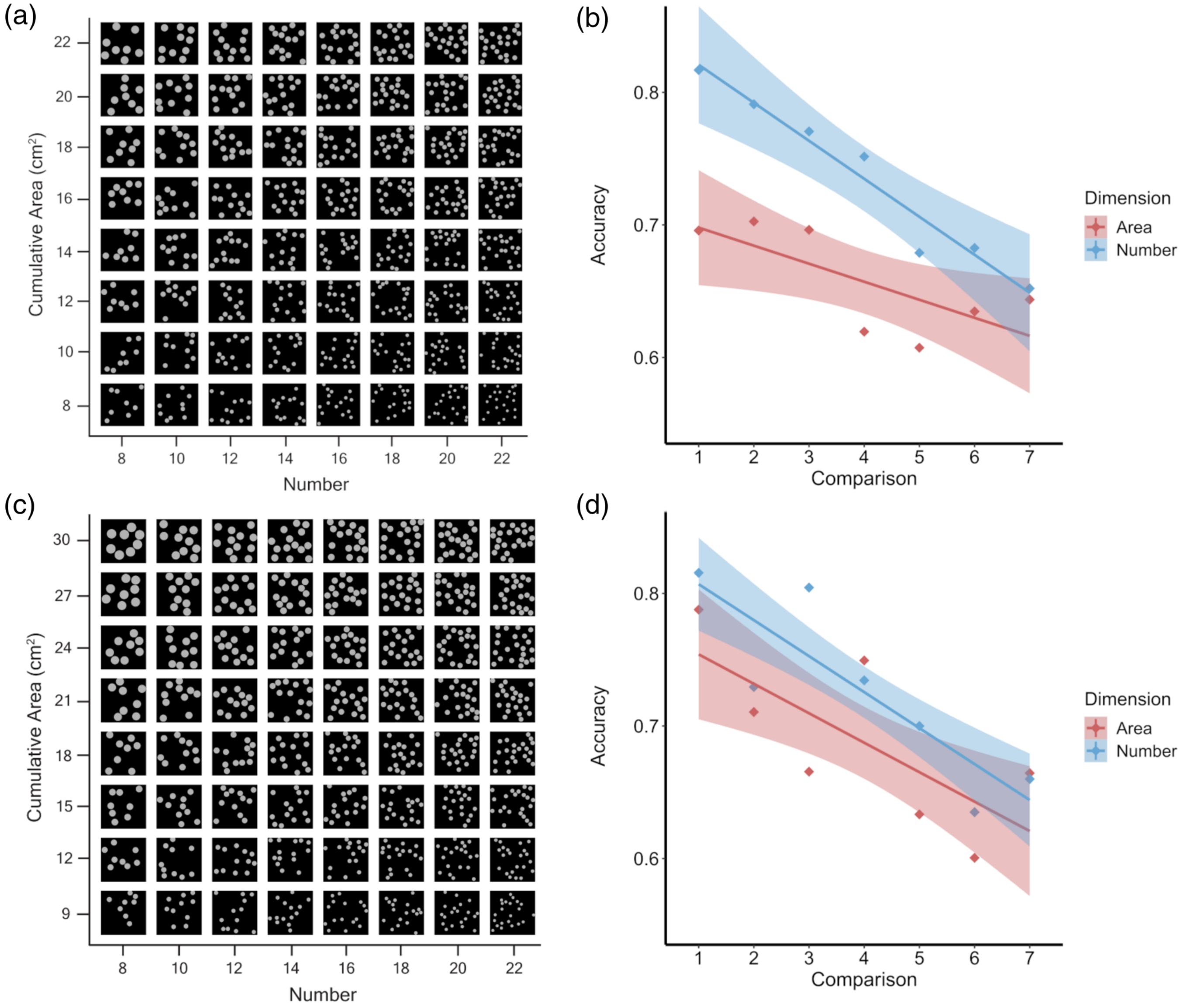

FIGURE 1 (**a and c**): The stimulus spaces used in Experiment 1, in which number and area were ratio-matched (**a**; Experiment 1A) or ratio-adjusted (**c**; Experiment 1B). (**b and d**): Adults' mean accuracy on the magnitude comparison tasks for each comparison with ratio-matched (**b**) and ratio-adjusted (**d**) stimuli. For example, for ratio-matched stimuli in Experiment 1A, comparison 1 refers to trials in which participants judged whether 8 or 10 dots was greater in number [blue], or whether 8 cm^2 or 10 cm^2 was greater in area [red]). Shaded areas represent the 95% confidence intervals

stimulus values (e.g., 8 and 10 dots) were compared. Trials involving non-adjacent stimulus values (e.g., 8 and 20 dots) were not analyzed. Accordingly, mean accuracy was evaluated as a function of 'comparison' (for an 8 × 8 stimulus space, 7 comparisons for each dimension). For example, 'comparison 1' refers to trials in which participants compared 8 and 10 dots (or 8 $cm^2$and 10 cm^2).

A repeated-measures ANOVA with the within-subjects factors of dimension (number or area) and stimulus comparison yielded a significant main effect of dimension on mean accuracy, $F(1, 17) = 9.72$, $p = .006$, $\eta_p^2 = .351$, suggesting that when dot arrays were matched on ratio, number was more perceptually discriminable than area (see Figure 1b). Moreover, there was a main effect of stimulus comparison (i.e., a ratio effect), $F(6, 108) = 5.50$, $p < .001$, $\eta_p^2 = .234$, but importantly, no dimension by comparison interaction, $F(6, 108) = .017$, $p = .367$, $\eta_p^2 = .058$, suggesting that the differential discriminability of number and area did not differ as a function of within-dimension discrimination difficulty.

2.2 | Ratio-adjusted stimuli

Following the results of Experiment 1A, in which adult participants exhibited greater accuracy for number judgments than area judgments, we developed a second stimulus set to account for this difference in accuracy (ratio-adjusted stimuli). Here, we uniformly increased the distance between area values by 1cm^2, such that area now differed by greater ratios than did number, with the prediction that this increase would result in equal discriminability. We evaluated the discriminability of number and area with these stimuli among adult participants, using a magnitude comparison task as in Experiment 1A, to determine whether number and area were in fact equally discriminable in these stimuli.

2.2.1 | Methods

Participants

Nineteenundergraduates (M_{age} = 20.48years) participated in this experiment for course credit. All participants had normal or corrected-to-normal vision. Procedures were approved by the local IRB. All tasks were presented on a desktop computer with a 19-inch screen (1280 × 1024px) located in an experimental laboratory. Participants sat approximately 60 cm from the computer screen.

Stimuli

Ratio-adjusted stimuli were identical to the stimuli used in Experiment 1A, except for the increased area ratios, implemented to account for differences in discriminability (see Figure 1c). In these ratio-adjusted stimuli, number values were 8, 10, 12, 14, 16, 18, 20, and 22dots, and area values were 9, 12, 15, 18, 21, 24, 27, and 30cm^2.

Procedure

The procedure was identical to that of Experiment 1A.

2.2.2 | Results

A repeated-measures ANOVA with the within-subjects factors of dimension (number and area) and stimulus comparison (7 comparisons for each dimension) found no effect of dimension on mean accuracy, $F(1, 18) = 2.41$, $p = .139$, $\eta_p^2 = .124$, suggesting no difference in discriminability for number and area with these stimuli (see Figure 1d). We further confirmed that there was no difference in discriminability for number and area using a Schuirmann's 'two one-sided tests' procedure, a test of equivalence (Schuirmann, 1987), $p < .001$ (Lakens et al., 2017). Moreover, there was a main effect of stimulus pair, $F(6, 102) = 9.51$, $p < .001$, $\eta_p^2 = .359$, but, importantly, no dimension by stimulus pair interaction, $F(6, 102) = 1.93$, $p = .083$, $\eta_p^2 = .102$, suggesting that the equal discriminability of number and area did not differ as a function of stimulus comparison. In other words, number and area were perceptually-matched across the stimulus space.

2.2.3 | Discussion

In Experiment 1, we found that when number and area were matched with respect to their ratios (e.g., 8 vs. 10 dots and 8 cm^{2}vs. 10 cm^2), adults were more accurate at discriminating stimuli on the basis of number than area. However, when area values were adjusted to reflect larger ratios than number (e.g., 8 vs. 10dots and 9cm^{2}vs. 12cm^2), adults were equally accurate at discriminating stimuli on the basis of number and area. These findings suggest that in previous work arguing for the unique salience of number, if number and area were matched by ratio, then this work may have confounded salience and discriminability.

3 | EXPERIMENT 2: MAGNITUDE COMPARISON (CHILDREN)

3.1 | Ratio-matched stimuli

In Experiment 2A, we evaluated the discriminability of number and area in ratio-matched stimuli using a magnitude comparison task with children. If number and area are not equally discriminable when matched by ratio, as was found for adults in Experiment 1A, then this would suggest that the unique salience of number previously observed in children (e.g., Ferrigno et al., 2017) could be an artifact of this difference in discriminability.

3.1.1 | Methods

Participants

Twenty children between 6 and 8 years of age (M_{age} = 6.92 years) participated in this experiment. All children received stickers throughout the session to maintain motivation, as well as a small gift at the end of the session for participating in the study. Caregivers provided written informed consent on behalf of their children. Procedures were approved by the local IRB. Children were tested in an experimental laboratory individually by an experimenter. All tasks were presented on a Hewlett Packard Compaq Elite 830023″ all-in-one desktop computer (resolution: 1920 × 1080 pixels). Children sat approximately 40 cm from the screen.

Stimuli

As in the previous experiment with adults, stimuli were dot arrays comprised of gray dots on a black background (400 × 400 px), varying parametrically in number and area. However, due to constraints on how many trials children can complete while remaining attentive and motivated, here we used reduced stimulus spaces (4 × 4). This 4 × 4 stimulus space reflects every other row/column of the full 8 × 8 stimulus spaces described in Experiment 1A (see Figure 2a). Thus, number and area values were 8, 12, 16, and 20 dots, and 8, 12, 16, and 20 cm^2, respectively.

Procedure

Children completed four blocks of trials (two number blocks and two area blocks; counterbalanced order). In the number blocks, children were asked to indicate which of two dot arrays was greater in number; in the area blocks, children were asked to indicate which of two dot arrays was greater in area. Each block consisted of 24 trials (96 total trials; 48 number trials and 48 area trials). Trials were randomly selected, without replacement, from all possible stimulus pairs in which the stimuli differed in both number and area ($n = 144$).

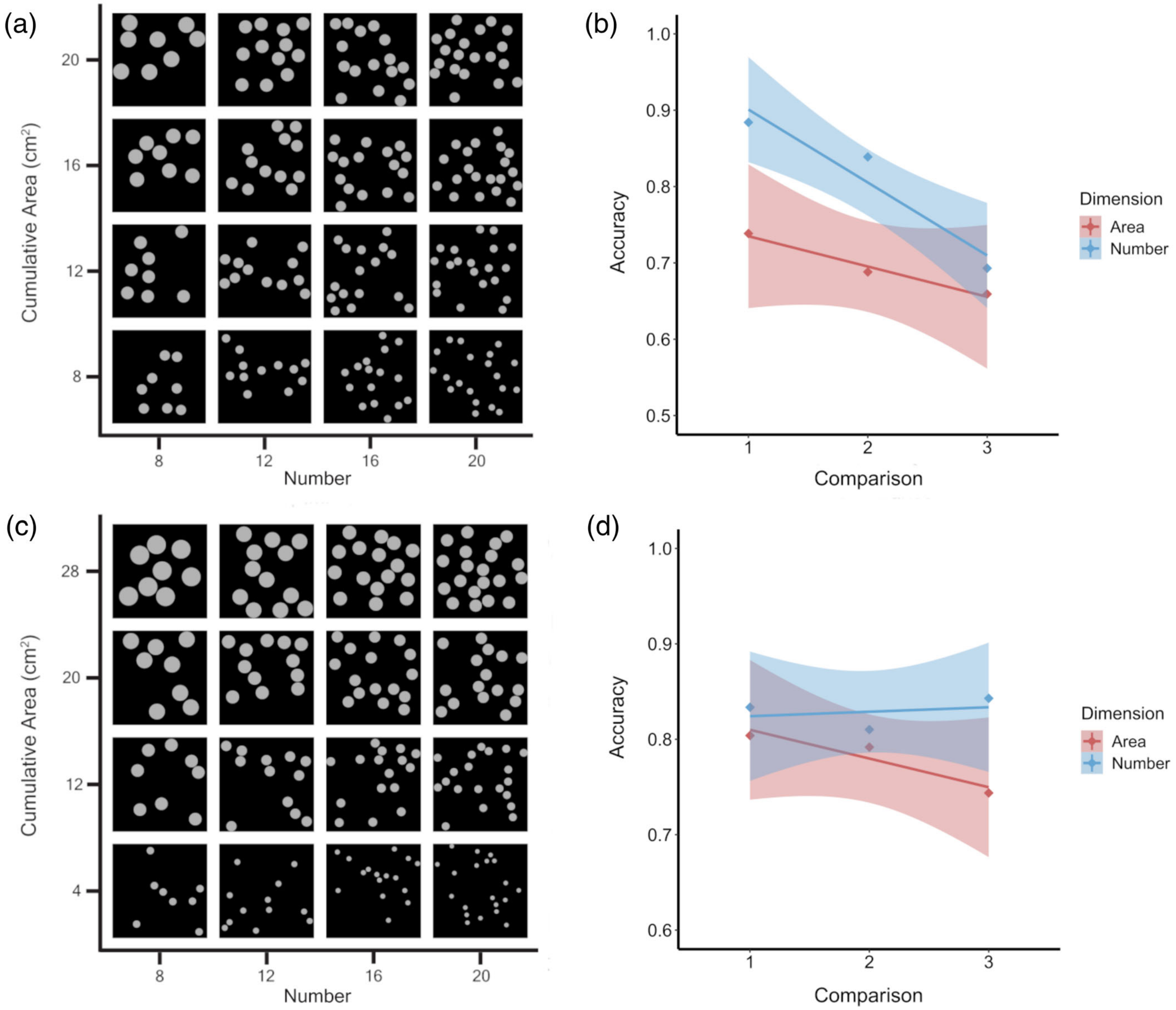

FIGURE 2 (**a and c**): The stimulus spaces used in Experiment 2, in which number and area were ratio-matched (**a**; Experiment 2A) or ratio-adjusted (**c**; Experiment 2B). (**b and d**): Children's mean accuracy on the magnitude comparison tasks for each comparison with the ratio-matched (**b**) and ratio-adjusted (**d**) stimuli. For example, for the ratio-matched stimuli in Experiment 2A, comparison 1 refers to trials in which participants judged whether 8 or 12 dots was greater in number [blue], or whether 8 cm^2 or 12 cm^2 was greater in area [red]). Shaded areas represent the 95% confidence intervals

Stimuli were presented on the center left and center right of the screen (side counterbalanced across trials), for 1500 ms (after which they were occluded). Children were instructed to tap the stimulus (left or right) that was greater in number (or area). Children had unlimited time to respond.

3.1.2 | Results

As in Experiment 1, to determine whether number and area were equally discriminable, we evaluated children's accuracy on trials in which adjacent stimulus values (e.g., 8 and 12 dots) were compared. Trials involving non-adjacent stimulus values (e.g., 8 and 20 dots) were not analyzed. Accordingly, mean accuracy was evaluated as a function of 'comparison' (for a 4×4 stimulus space, 3 comparisons for each dimension). For example, 'comparison 1′ refers to trials in which participants compared 8 and 12 dots (or 8 $cm^2$and 12 cm^2).

A repeated-measures ANOVA with the within-subjects factors of dimension (number or area) and comparison (3 comparisons for each dimension) yielded a significant main effect of dimension on mean accuracy, $F(1, 19) = 6.11$ $p = .023$, $\eta_p^2 = .243$, suggesting that when dot arrays were matched on ratio, number was more perceptually discriminable than area (see Figure 2b). Moreover, there was a main effect of comparison (i.e., a ratio effect), $F(2, 38) = 8.88$, $p < .001$, $\eta_p^2 = .219$, but importantly, no dimension by comparison interaction, $F(2, 38) = 1.94$, $p = .157$, $\eta_p^2 = .093$, suggesting that the differential discriminability of number and area did not differ as a function of within-dimension discrimination difficulty.

3.2 | Ratio-adjusted stimuli

As with the adult participants in Experiment 1B, we developed a second stimulus set to account for the difference in accuracy when comparing number and area. We uniformly increased the distance between area values by 2 cm^2, such that area now differed by greater ratios than did number, with the prediction that this increase would result in equal discriminability for children. In Experiment 2B, we evaluated the discriminability of number and area with these stimuli, using a magnitude comparison task, as in Experiment 2A, to determine whether number and area were in fact equally discriminable in these stimuli.

3.2.1 | Methods

Participants

Twenty children between 6and 8 years of age ($M_{age} = 7.23$ years) participated in this experiment. All children received stickers throughout the session to maintain motivation, as well as a small gift at the end of the session for participating in the study. Caregivers provided written informed consent on behalf of their children. Procedures were approved by the local IRB. Children were tested in an experimental laboratory individually by an experimenter. All tasks were presented on a Hewlett Packard Compaq Elite 830023″ all-in-one desktop computer (resolution: 1920 × 1080 pixels). Children sat approximately 40 cm from the screen.

Stimuli

Stimuli were identical to those of the previous experiments, except for the adjustment to the area ratios, implemented in order to account for differences in discriminability (see Figure 2c). This resulted in an 8 × 8 stimulus space with number values of 8, 10, 12, 14, 16, 18, 20 and 22, and area values of 4, 8, 12, 16, 20, 24, 28, and 32 cm^2. As in Experiment 2A, we reduced this to a 4 × 4stimulus space that reflected every other row/column of the full 8 × 8 stimulus space. Thus, in the 4 × 4 perceptually-matched stimulus space, number values were 8, 12, 16, and 20, and area values were 4, 12, 20, and 28 cm^2. For each stimulus set, five exemplars were generated for each pair of stimulus values (random dot placement), resulting in 80 unique stimuli per set (16 stimuli × 5 exemplars).

Procedure

The procedure was identical to that of Experiment 2A.

3.2.2 | Results

A repeated-measures ANOVA with the within-subjects factors of dimension (number or area) and stimulus comparison (3 comparisons for each dimension) found no effect of dimension on mean accuracy, $F(1, 19) = .331$ $p = .572$, $\eta_p^2 = .017$, suggesting no difference in discriminability for number and area for these stimuli (see Figure 2d). We further confirmed that there was no difference in discriminability for number and area using a Schuirmann's 'two one-sided tests' procedure (Schuirmann, 1987), $p = .001$ (Lakens, 2017). There was no main effect of comparison, $F(2, 38) = 2.01$, $p = .148$, $\eta_p^2 = .096$, and no dimension by comparison interaction, $F(2, 38) = 1.68$, $p = .201$, $\eta_p^2 = .081$, suggesting that the equal discriminability of number and area did not differ as a function of comparison. In other words, number and area were perceptually-matched across the stimulus space.

3.2.3 | Discussion

In Experiment 2, we found that when number and area were matched with respect to their ratios (e.g., 8 vs. 12 dots and 8 cm^{2}vs. 12 cm^2), children, like adults (Experiment 1), were more accurate at discriminating stimuli on the basis of number than area. However, when area values were adjusted so as to reflect larger ratios than number (e.g., 8 vs. 12 dots and 4 cm^{2}vs. 12 cm^2), children were equally accurate at discriminating stimuli on the basis of number and area. These findings suggest that in previous work arguing for the unique salience of number, if number and area were matched by ratio, then this work may have confounded salience and discriminability.

4 | EXPERIMENT 3: CATEGORIZATION (ADULTS)

In Experiment 3, we evaluated the relative salience of number and area during spontaneous categorization, for ratio-matched and perceptually-matched stimuli. If number is uniquely salient to adult participants, then participants should preferentially categorize stimuli by number for both the ratio-matched and perceptually-matched stimuli.

4.1 | Methods

4.1.1 | Participants

Forty-four undergraduates ($n = 22$ / condition; ratio-matched condition: $M_{age} = 19.68$ years; perceptually-matched condition: $M_{age} = 19.51$ years) participated in this experiment for course credit.

4.1.2 | Stimuli

Stimuli were identical to that of Experiment 1. The 'ratio-adjusted' stimuli described in Experiment 1B are here referred to as 'perceptually-matched' stimuli, given our findings that number and area were equally discriminable to adult participants in those stimuli.

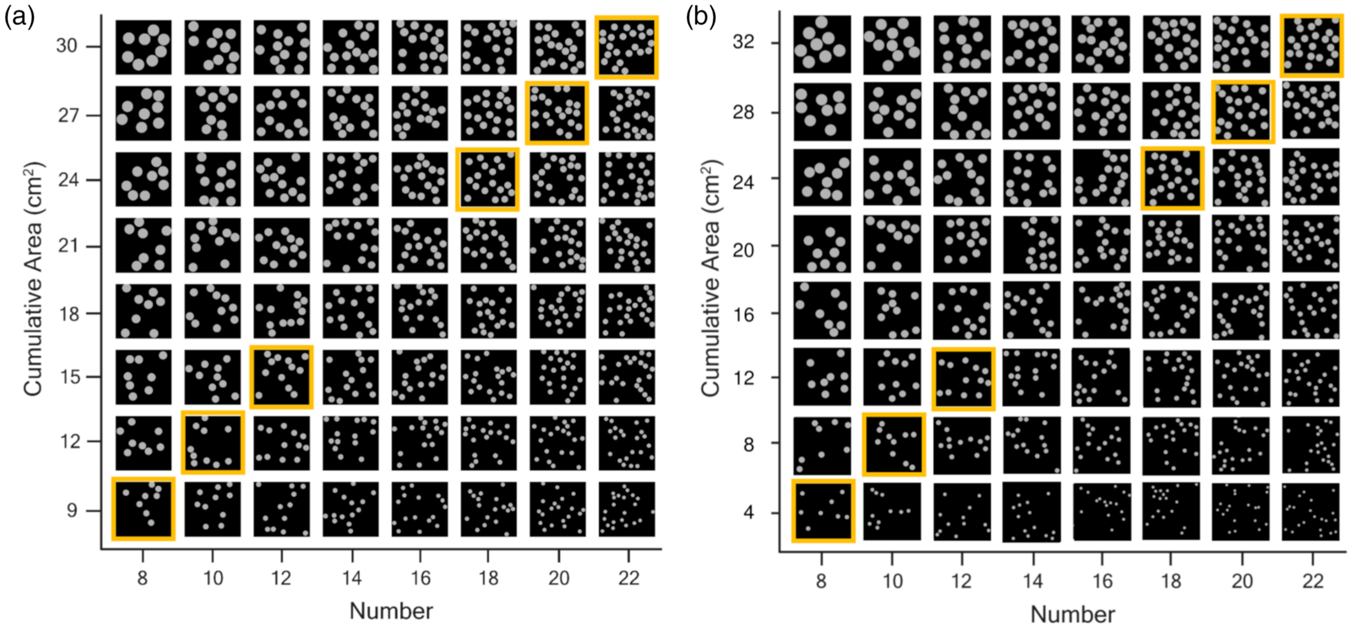

FIGURE 3 The perceptually-matched stimulus spaces used with adults in Experiment 3 (**a**) and children in Experiment 4 (**b**). Stimuli outlined in yellow depict the training categories (bottom left: 'small' category; top right: 'large' category)

4.1.3 | Procedures

Procedures were approved by the local IRB. All tasks were presented on a desktop computer with a 19-inch screen (1280 × 1024 px) located in an experimental laboratory. Participants sat approximately 60 cm from the computer screen.

In this categorization task, participants first completed a training phase, in which they learned to categorize stimuli into two groups using the 'q' and 'p' keys (counterbalanced across participants) and received corrective feedback.

In the training trials, number and area were correlated, with one category consisting of 'small' stimulus values (ratio-matched: 8, 10 and 12 dots, and 8, 10 and 12 cm^2; perceptually-matched: 8, 10 and 12 dots, and 9, 12 and 15 cm^2) and the other category consisting of 'large' stimulus values (ratio-matched: 18, 20 and 22 dots, and 18, 20 and 22 cm^2; perceptually-matched: 18, 20 and 22 dots, and 24, 27 and 30 cm^2; see Figure 3a). Participants completed 30 training trials (6 training stimuli × 5 exemplars). Critically, because number and area were fully correlated, participants could successfully complete training by categorizing stimuli based on number alone, area alone, or both dimensions. Participants were not given verbal category labels of any kind.

Participants then completed a test phase. As in the training phase, participants categorized stimuli using the 'q' and 'p' keys. Participants completed 302 trials. Participants were instructed to continue to categorize stimuli as they did during training, but that they would no longer receive feedback. Critically, number and area were *uncorrelated* during the test phase, allowing for an assessment of the unique contributions of number and area to category judgments.

In both the training and test phases, a single stimulus was presented centrally on each trial. The stimulus remained on-screen and visible until participants made a response (as in Ferrigno et al., 2017).

4.2 | Results

Mean accuracy during training trials indicated that participants successfully learned to categorize the stimuli (ratio-matched condition: $M = .922$; perceptually-matched condition: $M = .924$). To evaluate performance during test trials, we calculated the proportion of trials on which each stimulus was categorized into the 'large' group, for each participant (see Figures 4a,c). As in Ferrigno et al. (2017), we evaluated the contributions of number and area to participants' category judgments with a mixed-effects logistic regression model[2]. This model assessed category choice at the group level, with number and area as predictors (fixed effects) and with random slopes and intercepts for participants.

In the ratio-matched condition, the mixed-effects logistic regression yielded significant effects of both number ($\beta = 0.52$, $p < .001$) and area ($\beta = 0.13$, $p < .001$). However, a direct comparison (z-test) of the number and area beta weights revealed that the influence of number on category choice was significantly greater than the influence of area, $z = 6.11$, $p < .001$ (see Figure 4b).

In the perceptually-matched condition, the mixed-effects logistic regression yielded significant effects of both number ($\beta = 0.41$, $p < .001$) and area ($\beta = 0.20$, $p < .001$), as in the ratio-matched condition. Moreover, a direct comparison (z-test) of the number and area beta weights revealed that the influence of number on category choice was significantly greater than the influence of area, $z = 4.02$, $p < .001$, again, as in the ratio-matched condition (see Figure 4d).

To better understand the robustness of these effects, we then evaluated each participant's responses separately using a logistic regression model. Analyses of these individual differences revealed that, in both conditions, the number beta weights were greater than the area beta weights for the majority of participants (ratio-matched condition:

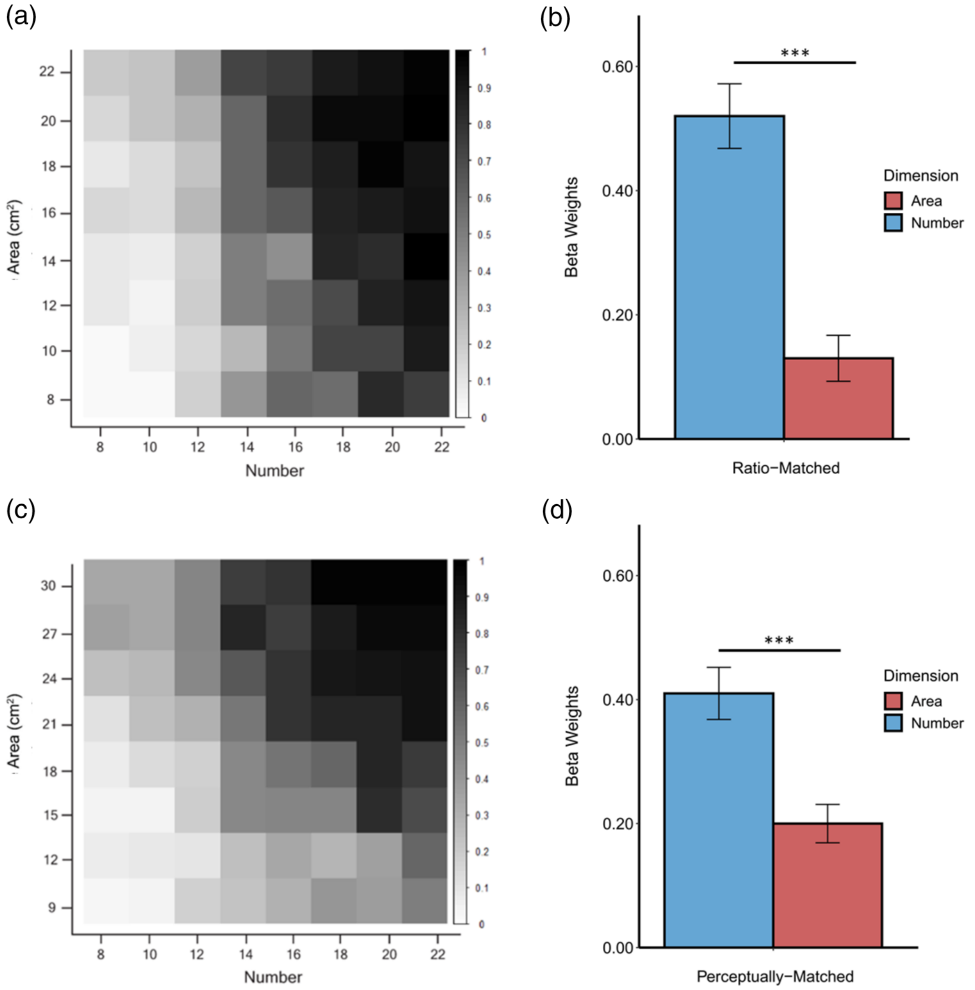

FIGURE 4 **(a and c)**: Mean proportion of trials in which adults categorized each stimulus into the 'large' category for ratio-matched (**a**) and perceptually-matched (**c**) stimuli. Each cell in the matrix corresponds to the stimuli at that location in the stimulus space. (**b and d**): Mean beta weights for the influence of number and area on adults' category judgments for ratio-matched (**b**) and perceptually-matched (**d**) stimuli. Error bars represent +/- 1 SE. ***$p < .001$

20/22; binomial test, $p < .001$; perceptually-matched condition: 17/22; binomial test, $p = .018$), demonstrating consistency across participants.

But could participants' categorization behavior instead reflect a reliance on other non-numerical dimensions, such as density? Reliance on density, a non-numerical dimension previously implicated in number perception (Dakin et al., 2011; Durgin, 1995), could result in successful performance during training. That is, the training categories of small number/area and large number/area additionally correspond to low and high density, respectively.

To evaluate the role of density (calculated as area / convex hull) in participants' behavior, we conducted group-level mixed-effects models with number, area, and density as predictors[3] (fixed effects) and with random slopes and intercepts for participants. In the ratio-matched condition, the mixed-effects logistic regression yielded significant effects of number ($\beta = 0.52$, $p < .001$) and area ($\beta = 0.13$, $p = .006$), as before, but not density ($\beta = 0.04$, $p = .757$). In the perceptually-matched condition, the mixed-effects logistic regression yielded significant effects for all dimensions: number ($\beta = 0.37$, $p < .001$), area ($\beta = 0.17$, $p < .001$), and density ($\beta = 0.35$, $p = .006$). Across both conditions, number beta values were significantly larger than area beta values, *ps* $< .001$, suggesting a bias for number over area. In the perceptually-matched condition, however, density beta values did not differ from number or area beta values, *ps* $> .156$. These results suggest that density may have been used in addition to number, but it is unclear to what extent it might be favored compared to the other dimensions. Moreover, because density was neither ratio-matched nor perceptually-matched to number and area, these results should be interpreted with caution, as it is unclear to what extent the role of density in categorization may be due to differences in discriminability.

4.3 | Discussion

The findings of Experiment 3 suggest that adults' number bias during categorization does not simply reflect differences in perceptual discriminability. That is, even when number and area were equally discriminable, adults' categorization choices were influenced by number significantly more than area, providing strong support for the greater salience of number compared to area, at least by adulthood. Altogether, these findings replicate existing work in which there is a bias for number over area in adults, and extend these findings by showing that this bias is not simply due to a mismatch in discriminability (see also, Castaldi et al., 2018). Nevertheless, it remains an open question how number and area compare to other co-occurring dimensions such as density.

5 | EXPERIMENT 4: CATEGORIZATION (CHILDREN)

One possible interpretation of the findings of Experiment 3 is that the number bias is innate. However, other research suggests that math education may lead to an increased number bias (Braham et al., 2018; Piazza et al., 2018). Accordingly, this finding raises an important question regarding the developmental trajectory of number and area perception. With perceptually-matched stimuli, do children, like adults, exhibit a significant number bias, consistent with the Number Sense account? Or, do children exhibit a significant area bias, suggesting a dominant role of non-numerical magnitude information earlier in development, contra the Number Sense account (Gabay et al., 2013; Leibovich et al., 2017; Newcombe et al., 2015)?

To this end, in Experiment 4, we evaluated the relative salience of number and area during children's spontaneous categorization, for ratio-matched and perceptually-matched stimuli. If number is intrinsically salient to children, then participants should preferentially categorize stimuli by number for both the ratio-matched and perceptually-matched stimuli.

5.1 | Methods

5.1.1 | Participants

Forty-four children between 6 and 8 years of age (n = 22 / condition; ratio-matched condition: M_{age} = 6.70 years; perceptually-matched condition: M_{age} = 6.96 years) participated in this experiment. All children received stickers throughout the session to maintain motivation, as well as a small gift at the end of the session for participating in the study. Caregivers provided written informed consent on behalf of their children. Procedures were approved by the local IRB. Children were tested in an experimental laboratory individually by an experimenter. All tasks were presented on a Hewlett Packard Compaq Elite 830023″ all-in-one desktop computer (resolution: 1920 × 1080 pixels). Children sat approximately 40 cm from the screen.

5.1.2 | Stimuli

Ratio-matched stimuli were identical to those used in Experiments 1A and 3. This 8 × 8 ratio-matched stimulus space reflects an expanded version of the 4 × 4 stimulus space used in Experiment 2A. Perceptually-matched stimuli reflected an expanded version (8 × 8) of the 4 × 4 perceptually-matched stimulus space used in Experiment 2B (see Figure 3b). Number values were 8, 10, 12, 14, 16, 18, 20 and 22 dots, and area values were 4, 8, 12, 16, 20, 24, 28 and 32 cm^2.

5.1.3 | Procedure

As in Experiment 3, we used a spontaneous categorization task to evaluate the relative salience of number and area. In this task, children first completed a training phase, in which they learned to categorize stimuli into two groups denoted by two arbitrary onscreen buttons (yellow star and purple diamond), presented on the center left and center right of the screen (side counterbalanced across participants), as in Ferrigno et al. (2017). Children received corrective feedback.

In the training phase, number and area were correlated, with one category consisting of 'small' stimulus values (ratio-matched: 8, 10 and 12 dots, and 8, 10 and 12 cm^2; perceptually-matched: 8, 10 and 12 dots, and 4, 8 and 12 cm^2) and the other category consisting of 'large' stimulus values (ratio-matched: 18, 20 and 22 dots, and 18, 20 and 22 cm^2; perceptually-matched: 18, 20 and 22 dots, and 24, 28 and 32 cm^2; see Figure 3b). Children completed 30 training trials (6 training stimuli × 5 exemplars). No category labels were provided.

Children then completed a test phase. The test phase was separated into four blocks (21 trials / block; 84 total trials). Children were instructed to continue to categorize stimuli as during training but that they would no longer receive feedback.

In both the training and test phases, stimuli were presented centrally and remained on-screen until children responded. Children initiated their response by first tapping the stimulus. After tapping the stimulus, the stimulus disappeared, and the two category icons (yellow star and purple diamond) appeared on the center-left and center-right of the screen. To categorize the stimulus, children then tapped one of the two category icons. They had unlimited time to respond.

5.2 | Results

Mean accuracy during training trials indicated that children successfully learned to categorize the stimuli (ratio-matched condition: $M = .895$; perceptually-matched condition: $M = .897$). To evaluate performance during test trials, we calculated the proportion of trials on which each stimulus was categorized into the 'large' group, for each child (see Figures 5a,c). As in Experiment 3, we evaluated the contributions of number and area to participants' category judgments with a mixed-effects logistic regression model. This model assessed category choice at the group level, with number and area as predictors (fixed effects) and with random slopes and intercepts for participants.

In the ratio-matched condition, the mixed-effects logistic regression model yielded significant effects of both number ($\beta = 0.61$, $p < .001$) and area ($\beta = -0.08$, $p = .011$). However, a direct comparison (z-test) of the number and area beta weights found that the influence of number on

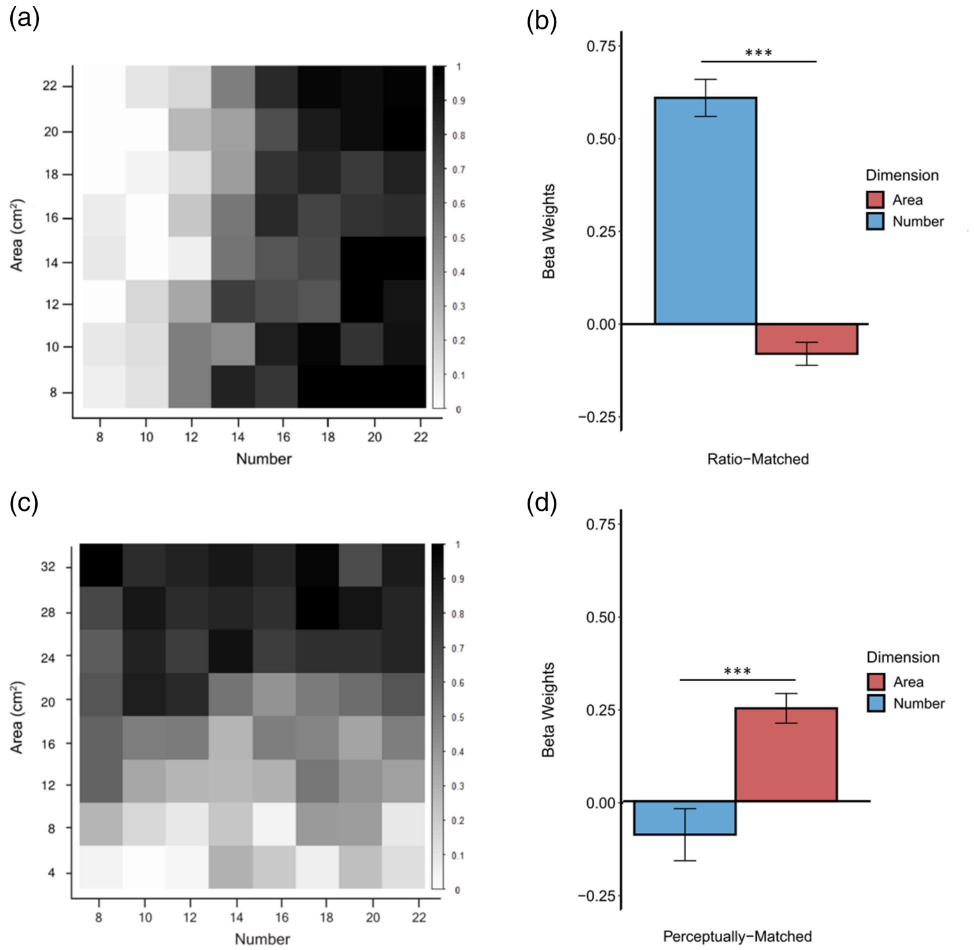

FIGURE 5 **(a and c)**: Mean proportion of trials in which children categorized each stimulus into the 'large' category for ratio-matched (**a**) and perceptually-matched (**c**) stimuli. Each cell in the matrix corresponds to the stimuli at that location in the stimulus space. (**b and d)**: Mean beta weights for the influence of number and area on children's category judgments for ratio-matched (**b**) and perceptually-matched (**d**) stimuli. Error bars represent +/- 1 SE. ***$p < .001$

category choice was significantly greater than the influence of area, $z = 11.73$, $p < .001$ (see Figure 5b).

In the perceptually-matched condition, the mixed-effects logistic regression model yielded a significant effect of area ($\beta = 0.25$, $p < .001$), but not number ($\beta = -0.09$, $p = .176$). Moreover, a direct comparison (z-test) of the number and area beta weights found that the influence of area on category choice was significantly greater than the influence of number, $z = 4.22$, $p < .001$ (see Figure 5d).

As in Experiment 3, we also evaluated each participant's responses separately using a logistic regression model, to better understand the robustness of the effects. Analyses of these individual differences revealed that, in the ratio-matched condition, the number beta weights were greater than the area beta weights for all participants (22/22; binomial test, $p < .001$), whereas in the perceptually-matched condition, the area beta weights were greater than the number beta weights for the majority of participants (17/22; binomial test, $p = .018$).

We also conducted exploratory analyses to evaluate the potential influence of density on children's category judgments[4]. And as in Experiment 3, these analyses should be interpreted with caution given that density was not ratio-matched (or perceptually-matched) to the other dimensions. In the ratio-matched condition, the mixed-effects logistic regression yielded a significant effect of number ($\beta = 0.63$, $p < .001$), but not area ($\beta = -0.05$, $p = .187$) or density ($\beta = -0.29$, $p = .059$). In the perceptually-matched condition, the mixed-effects logistic regression yielded a significant effect of area ($\beta = 0.25$, $p < .001$), but not number ($\beta = -0.10$, $p = .145$) or density ($\beta = 0.10$, $p = .638$). Thus, in both conditions, density did not appear to influence children's category judgments.

However, another possibility is that element size may have influenced children's category judgments. Specifically, due to the increases in area required to create perceptually-matched stimuli in children, element size necessarily varied across training stimuli, such that stimuli in the 'small' category had smaller-sized dots than the 'large' category (see Figure 3b). Thus, it is possible that element size may have played a role in children's category judgments, which would be consistent with previous work showing greater interference by element size on number judgments in children (Soltész et al., 2010) and adults with dyscalculia (Castaldi et al., 2018) compared to typical adults.

To this end, we conducted exploratory analyses to evaluate the potential influence of element size on children's category judgments in the perceptually-matched condition[5]. Given that the present results suggest a primary role of area in children's category judgments, we ruled out the effect of element size by comparing the effects of element size and area when both were predictors of children's category judgments. A mixed effects logistic regression model with element size and area as predictors (fixed effects) and random intercepts for participants was conducted. This analysis yielded a significant effect of area ($\beta = 0.18$, $p < .001$), but not element size ($\beta = 0.09$, $p = .366$), suggesting that despite differences in element size across training categories, children's categorization behavior was nonetheless predominantly influenced by area during test.

5.3 | Discussion

The results of Experiment 4 suggest that when number and area were equivalent in discriminability, children categorized the stimuli primarily based on area, not number. Children's categorization behavior stands in contrast to that of adults (Experiment 3), whose categorization behavior was influenced by number more than area, regardless of whether stimuli were ratio- or perceptually-matched. Contra other recent work (Ferrigno et al., 2017; Tomlinson et al., 2020), this finding suggests that there is no intrinsic number bias during childhood.

Moreover, although the stimulus spaces used here were not perceptually matched with respect to density and element size, exploratory analyses suggest that children's category judgments were not significantly influenced by density or element size. Critically, however, future work should evaluate this hypothesis with stimuli where density and element size are also perceptually matched, to ensure that the present results were not driven by unequal discriminability among a subset of dimensions.

6 | GENERAL DISCUSSION

The findings of the present study suggest that although number is more salient than non-numerical magnitudes by adulthood, number may not be intrinsically more salient. First, and critically, we found that when number and area were matched by ratio, both adults (Experiment 1A) and children (Experiment 2A), exhibited greater acuity for number than area. That is, participants more easily discriminated stimuli by number than area. Importantly, though, by increasing the ratios in area relative to number, we generated stimuli in which number and area were equally discriminable in both adults and children (Experiments 1B and 2B, respectively). With these ratio- and perceptually-matched stimuli, we then asked whether the previously observed number bias differed as a function of stimulus condition. We found that adults preferentially categorized by number, regardless of stimulus condition (Experiment 3), suggesting that by adulthood, number is uniquely salient, and that this salience exists even in the absence of explicit magnitude judgments.

In contrast, we found that children's categorization preferences differed by stimulus condition, with children preferentially categorizing by number when stimuli were ratio-matched but by area when stimuli were perceptually-matched (Experiment 4). These findings suggest that the number bias previously observed in children may have resulted from a mismatch in perceptual discriminability for the different dimensions, *not* the intrinsic salience of number. When this mismatch in discriminability is accounted for, area is more salient than number during childhood. Furthermore, in contrast to adults (Experiment 3), the present results suggest that when stimuli are perceptually matched for number and area, children exhibit a bias for area over number, and perhaps also density and element size, suggesting that not all non-numerical magnitudes are equally salient. Taken together, these findings suggest that, contra previous work, number is not an innately salient dimension but, rather, a bias towards number emerges over development.

These findings present a significant challenge to a key claim of the Number Sense account; namely, that number is a naturally dominant dimension (Cicchini et al., 2016; Cordes & Brannon, 2008; Ferrigno et al., 2017). Specifically, the present work suggests that the number bias children have exhibited in past work (and, here, with ratio-matched stimuli) reflects unequal discriminability, not the intrinsic salience of number. However, we found that there is developmental change, such that by adulthood, number is uniquely salient, consistent with previous work, even when stimuli are perceptually-matched and when no explicit magnitude judgment is required.

Evidence of an area bias in children suggests that claims of numerical salience in infancy and early childhood may have resulted from a conflation between acuity and salience. Importantly, we do not dispute the findings that humans, across development, exhibit better acuity for number than cumulative area (Cordes & Brannon, 2008; Lourenco & Bonny, 2017; Odic, 2018). Infants and older children alike show greater acuity for number compared to area (Brannon et al., 2004; Tomlinson et al., 2020; cf. Odic et al., 2013). Our findings in Experiments 1and 2 of the present work likewise demonstrate greater acuity for number than area (for discussion of potential explanations for this difference in acuity, see Aulet & Lourenco, 2021a). We disagree, however, that this difference in acuity indicates an intrinsic salience of number. Rather, it is precisely this difference in acuity that thus far has undermined the ability to assess the relative salience of magnitude dimensions. Although we did not test young children or infants in the present work, the area bias in children (and number bias in adults) observed here is consistent with a developmental shift in magnitude salience from non-numerical to numerical magnitudes (Hamamouche & Cordes, 2019; Newcombe, 2014). Accordingly, such a shift predicts that younger children and infants would exhibit as much of an area bias, if not more, than the children in the present work.

One might argue that this differential acuity is itself evidence for the primacy of number, positing that if number and area were equally important to the observer, then acuity would likewise be equal. This assumes, however, that the mathematical instantiations of the dimensions reflect the perceptual reality. Though rarely challenged, such an assumption is not a logical necessity (Barth, 2008). Indeed, recent work on the 'additive-area heuristic' suggests that area perception may be better characterized by the sum of the element dimensions (e.g., length + width), rather than mathematical area (Yousif & Keil, 2019).

Thus, one possible explanation for the observed differences in discriminability is that we have thus far been incorrect about the perceptually relevant unit for area perception (see also, Barth, 2008). Given this debate over computations underlying area perception, we chose to perceptually-match area to number in a data-driven manner, in order to remain agnostic as to the correct area model. In the present work (see also, Aulet & Lourenco, 2021b), number and area were perceptually-matched at the group level, but in future work, it may prove fruitful to perceptually-match more than two dimensions, and at the individual subject level, given known variability in magnitude acuity across dimensions and across individuals (DeWind et al., 2015; Lourenco et al., 2012; Starr et al., 2017). Although the perceptual matching of multiple dimensions may require more work upfront when designing stimuli, it is a necessary precondition for adjudicating between the different perspectives within the study of magnitude perception.

Nevertheless, it is unclear whether the Magnitude Sense account, which posits that non-numerical magnitudes are more salient than number, can explain the present results. In particular, this account argues that inhibitory control is required, even in adulthood, to discriminate number, because non-numerical magnitudes are processed more automatically than number and, thus, must be inhibited (Leibovich et al., 2017). Thus, it is unclear whether this account can explain why adults preferentially categorized the stimuli by number, even when number and area were uncorrelated, no explicit numerical judgment was required, and the task could be successfully completed using non-numerical magnitude (i.e., area). Accordingly, though the present findings provide evidence against the Number Sense account, they are not necessarily fully consistent or compatible with the Magnitude Sense account. These incompatibilities point to a need for theories of number perception that can explain both the non-numerical bias early in life, as well as the numerical bias in adulthood (Lourenco & Aulet, 2022).

When number and area were perceptually-matched, we also found that adults, but not children, were significantly influenced by the density of the dot arrays. Of note, however, is that the stimuli in the present work were designed to evaluate the role of number and area, specifically. As a result, density was not directly manipulated and was not explicitly perceptually-matched to number and area. Thus, the relative influence of density may not reflect its intrinsic salience compared to number and area. Despite this caveat, it is interesting that, whereas adults might make use of number and density, children were consistent in their exclusive use of area (in the perceptually-matched condition), perhaps reflecting the relative ease of unidimensional categorization (Ashby et al., 1999). Accordingly, the differences between children and adults observed here may reflect adults' greater sensitivity to the statistical relations between co-occurring magnitude dimensions. Although such an interpretation is speculative, future research could address it directly by accounting for potential differences in discriminability between density and other dimensions such as number and area.

As discussed in Experiment 4, the increased ratio differences that were required to perceptually-match number and area in children resulted in a systematic difference in element size during categorization training. That is, training stimuli belonging to the 'small' category were smaller than the 'large' category, in number, area, and also element size, making it possible that children used element size to guide their category judgments. Although we were unable to simultaneously evaluate the effects of number, area, and element size due to issues of high multicollinearity (Park, 2021), we found that when only area and element size were included as predictors, area, but not element size, predicted children's category judgments. Nonetheless, it is possible that even though children were not significantly influenced by element size during test trials, element size differences during training may have predisposed children to attend to all non-numerical information, more generally, during test trials. However, this seems unlikely given that we found an influence of area, but not density or element size, on category judgments. If element size differences during training biased children towards non-numerical magnitudes generally, then one would expect an increase in the influence of all non-numerical magnitudes, including density and element size. We included analyses to rule out this possibility, showing that when area and element size were used as predictors of category judgments, area, but not element size, predicted category judgments. Accordingly, the more parsimonious interpretation of the present results is that area was most salient during training trials, and thus, was also most salient during test trials.

One key limitation of the present work is the use of a task with unlimited stimulus presentation time. In the spontaneous categorization task (Experiments 3 and 4), stimuli remained onscreen until participants made their response (i.e., selected which category the stimulus belonged to). Importantly, the spontaneous categorization task used here was based on the work of Ferrigno et al. (2017). Because Ferrigno et al. allowed unlimited time for participants' responses, we likewise allowed unlimited time for participants' responses. However, the unlimited presentation time may have affected the visual processing of the stimuli and/or influenced the strategies available to participants (see also, Leibovich-Raveh et al., 2018). For example, although human participants can discriminate number when stimuli are displayed for as little as 16 ms (Inglis & Gilmore, 2013), participant's acuity significantly increases with longer presentation times, potentially due to the role of serial accumulation in the formation of number representations (Cheyette & Piantadosi, 2019). Accordingly, it is possible the unlimited presentation time in the present study significantly influenced participants stimulus representations and resulting category judgments. Critically, however, presentation time was unlimited for both adults and children, and for both ratio-matched and perceptually-matched stimuli. Therefore, the differences in results across age and conditions cannot be explained by differences in presentation time.

Taken together, the current experiments present a significant challenge to the Number Sense account, which represents the dominant view of number perception (Anobile et al., 2016; Clarke & Beck, 2021). Most strikingly, the unique salience of area observed in childhood suggests that non-numerical magnitudes may be more dominant early in life (Henik et al., 2017; Viarouge et al., 2018), with number becoming increasingly salient over development. But an open question is: what accounts for the developmental shift in bias? Whereas children privilege area over number, adults privilege number over area. Piazza et al. (2018) provided evidence that the acquisition of symbolic number systems and exposure to formal mathematical education enhances our ability to focus on number, rather than non-numerical magnitudes (see also, Ferrigno et al., 2017; Piazza et al., 2013). These findings provide a potential mechanism for the developmental shift to a number bias, but not for why area is privileged in the first place. Relatedly, it is unclear whether non-human primates, or human populations that lack number experience, such as the Tsimané people (Ferrigno et al., 2017), would also privilege area over number when stimuli are perceptually-matched. If all groups, besides adults with extensive experience with symbolic numerals and formal math education, exhibit an area bias, this would suggest that natural selection favored the use of non-numerical information over numerical information. Such a possibility might be unsurprising given that non-numerical information (e.g., volume of food) is often more directly relevant to an animal's needs (e.g., calories) than numerical information (Agrillo & Bisazza, 2014).

Lastly, given that children can discriminate both number and area (as well as other cues), one might ask whether evaluating the relative salience of these dimensions is a productive research question. We would answer that it is. In addition to what dimensional salience can tell us about the perception of multidimensional stimuli across development, evaluating the relative salience of magnitude dimensions, specifically, may help to inform both typical and atypical mathematical development. For example, Castaldi et al. (2018) found that, in adults with dyscalculia, mean element size interfered with number judgments more than the reverse. This finding suggests that a failure to shift from a non-numerical bias to a numerical one over development may be indicative of difficulties learning formal mathematics (see also, Fuhs et al., 2021). By controlling for discriminability when assessing magnitude interactions, future work will be better equipped to dissociate the contributions of measurement error (Barth, 2008; Yousif & Keil, 2019), acuity (Aulet & Lourenco, 2021a), and intrinsic magnitude salience (Castaldi et al., 2018).

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CONFLICT OF INTEREST

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

The data from this study are available from the authors upon request.

ETHICS STATEMENT

Procedures were approved by the Institutional Review Board at Emory University.

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ENDNOTES

[1] Post hoc analyses found no main effect of block order in Experiments 1 and 2, *ps* > .208.

[2] For consistency with Ferrigno et al. (2017), group-level mixed-effects logistic regression models were used here; however, qualitatively similar results were observed when group-level logistic regression models (i.e., without random effects) were used.

[3] The Variance Inflation Factor (VIF) for models included in Experiment 3 were < 2, within the acceptable range (< 10; Hair, Anderson, Tatham, & Black, 1995), suggesting no issues of multicollinearity even when density was included in the model.

[4] The VIF for models reported in Experiment 4 were < 2, within the acceptable range (< 10; Hair et al., 1995), suggesting no issues of multicollinearity when density was included in the model.

[5] Due to high multicollinearity between the predictors of number, area, and element size (VIF > 10), we were unable to evaluate the influence of all three predictors concurrently.

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